CB-UC-1

Command Broker Use Case 1

Balancing Authority Operations — Integrated CB Interface

High-Tempo Decision Environment: Mixed Above- and Below-Bright-Line Action Management

Implementing Document to CB-Framework_Cross_Sector and CB-IB_Qualification_Standard

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For use by Gen AI system vendors, control system developers, bulk electric system operators, and balancing authority compliance personnel

Parent document reference: CB-Framework_Cross_Sector — CB-IB_Qualification_Standard

1. Purpose and Intended Use

This document is the first in the CB Use Case series. Use cases in this series translate the Command Broker (CB) architectural requirements established in the CB-Framework_Cross_Sector and the CB-IB_Qualification_Standard into vendor-facing interface specifications for defined deployment environments. Each use case describes a specific operational context, proposes the Bright Line placement appropriate to that context with the analytical basis stated, and specifies what a CB-compliant system interface must contain and how it must behave.

The primary audience for this use case series is the vendor and supplier community developing Gen AI-integrated control and advisory systems for deployment in critical infrastructure operations. Secondary audiences are the operators who will use these systems and the compliance personnel responsible for verifying that deployed systems satisfy CB requirements.

This use case describes the Balancing Authority (BA) operational environment: a high-tempo, continuous decision context in which a control room operator manages generation-to-load balance in real time, working with a Gen AI advisory system that presents multiple action options simultaneously. Some options will be above the Bright Line; others will be below it. The CB requirement applies to the above-Bright-Line options. The interface must handle both tiers in a single operational view without creating friction that undermines operational performance.

The interface architecture specified here is cross-sector by design. The BA scenario provides the illustrative context. Vendors adapting this specification for gas transmission control rooms, water system operations centers, or data center NOC environments substitute the relevant process parameters, consequence thresholds, and anomaly flag categories without changing the underlying interface architecture. Section 7 provides explicit guidance on adaptation for those environments.

The Bright Line placement proposed in this document reflects the author’s analytical judgment and is available for review and adjustment by subject matter experts with operational BA experience. The consequence analysis in Section 3 explicitly states the basis for each proposed placement, enabling reviewers to identify and challenge specific threshold determinations.

2. Operational Scenario: Balancing Authority Operations

2.1 Function and Responsibility

A Balancing Authority is the entity responsible for maintaining the instantaneous balance between electrical generation and load within a defined area of the bulk electric system (BES). The BA operator’s primary obligation, as defined in NERC Reliability Standard BAL-001, is to

maintain scheduled interchange with neighboring BAs while supporting frequency within the interconnection. In practice, this means the operator continuously monitors Area Control Error (ACE) — the real-time deviation between actual and scheduled net interchange, adjusted for frequency — and takes corrective action when ACE exceeds defined thresholds.

The BA operator works within a SCADA/EMS environment that displays real-time generation output, load, interchange, frequency, and ACE for the BA's footprint. A Gen AI advisory system operating in this environment analyzes the current system state. It presents the operator with a set of recommended actions — generation dispatch adjustments, interchange schedule modifications, or load management actions — ranked by the system's assessment of their effectiveness in restoring ACE to acceptable limits.

2.2 Decision Cadence and Tempo

The BA operational environment is characterized by continuous, high-frequency decision-making. ACE is computed every two to four seconds. Significant ACE deviations that require corrective action can develop within minutes. Generation dispatch instructions typically execute within seconds to minutes, depending on the resource type. The operator may be managing multiple simultaneous deviations, responding to unexpected unit trips, coordinating with neighboring BAs on emergency assistance, and monitoring developing situations — all within the same operational window.

A Gen AI advisory system in this environment may present updated action recommendations every thirty to sixty seconds as system conditions change. At any given moment, the operator may be evaluating a set of three to eight recommended actions that cover different resource combinations, dispatch magnitudes, and risk profiles. The interface must present this information in a form that supports rapid, informed decision-making without overwhelming the operator or creating decision fatigue.

2.3 Resource Classes in Scope

The BA dispatch environment involves multiple resource classes with materially different operational characteristics, response times, and consequence profiles. The following resource classes are relevant to Bright Line placement analysis:

- **Conventional thermal generation (coal, gas, nuclear).** Controlled through economic dispatch signals. Response times are measured in minutes. Dispatch adjustments are incremental and reversible. Technical specifications and reactor physics constrain nuclear unit dispatch.
- **Fast-response resources (gas peakers, batteries, demand response).** Response times range from seconds to minutes. High maneuverability. Batteries can transition from charging to discharging near-instantaneously. Demand response involves coordinated load reduction across customer classes.

- **Interchange schedules with neighboring BAs.** Modifications require coordination and may involve settling energy imbalances. Near-term schedule adjustments can be executed within the operator's authority; longer-horizon adjustments may require bilateral negotiation.
- **Transmission switching and topology changes.** Reconfiguring transmission paths to manage flows. These actions have system-wide effects that may not be immediately predictable and are not recoverable on short timescales if a contingency develops.
- **Emergency operations actions.** Load shedding, emergency energy requests, and automatic remedial action scheme (RAS) interactions. These are the highest-consequence actions with direct public impact and potential cascading reliability effects.

2.4 The Gen AI Advisory System in This Context

The Gen AI advisory system processes the real-time system state — ACE, generation output by unit, load by zone, interchange by path, frequency, and transmission flows — and generates a ranked set of recommended actions. For each recommended action, the system characterizes the expected ACE correction, the response time, the resource cost, and any identified constraints or anomalies. The operator evaluates these recommendations against their own situational awareness and selects the action or sequence of actions to execute.

The advisory system operates in continuous mode, updating recommendations as system conditions change. The operator does not wait for a recommendation before acting — they may act independently of the advisory system's current recommendation set if conditions require immediate response. The advisory system is a decision support tool, not a decision authority.

3. Bright Line Placement: Proposed Analysis

We propose the following Bright Line placement based on a consequence analysis of the BA operational environment. For each action category, the analysis identifies the nature of the harm if the Gen AI recommendation is erroneous and acted upon, the scale of the harm, its reversibility within the operational timeframe, and the tempo compatibility of a CB evaluation step. Placement follows the consequence of each action, as FD-BL §4.2 requires. It does not follow the technology that recommends the action, whether the recommendation is called advisory, or the layer in which the action originates. The proposed placement is presented for review and adjustment by BA operational subject matter experts.

The core principle governing placement: actions whose erroneous execution could produce a BES reliability event affecting multiple BAs, a public impact through involuntary load shedding, an irreversible system state change within the operator's response window, or a challenge to a

nuclear generating unit's technical-specification limits or reactor protection systems are above the Bright Line. Actions whose erroneous execution produces a correctable, facility-local consequence within the BA's direct control are below the Bright Line.

3.1 Proposed Below-Bright-Line Actions

The following action categories are proposed below the Bright Line. For each, the Gen AI system may present recommendations, and the operator may confirm execution of a defined action sequence with the two-step confirmation specified in R-BL-2. The CB disclosure element is not required; a simplified confirmation display is required as specified in Section 4.2.

Action Category	Consequence of Erroneous	Basis for Below-Bright-Line Placement
Incremental economic dispatch adjustment to online thermal units within the existing operating range	Sub-optimal fuel cost; minor ACE deviation persists briefly. Correctable within minutes by subsequent dispatch.	The consequence is economic, not reliability-affecting. Fully reversible. No public impact.
Incremental economic dispatch to a nuclear generating unit within its pre-agreed maneuvering band	Sub-optimal fuel cost; minor ACE deviation persists briefly. Correctable within minutes by subsequent dispatch. The unit remains inside the band its plant has agreed, away from technical-specification limits.	The band, set by the plant and enforced by its own controls rather than by the recommendation, bounds the consequence below the threshold. Same basis as incremental thermal dispatch. Reversible. No public impact. The placement follows the instruction's consequence, not the unit's fuel type.
Battery storage charge/discharge dispatch within the battery controller's operating limits	Non-optimal state-of-charge management. ACE correction may be suboptimal. Correctable within the operating hours.	The battery controller's operating limits bound the consequence, and the bound is enforced by those limits rather than by the recommendation. Being placed below the line, the dispatch needs no authorization, in advance or in the moment, and the limits are an operating band rather than a pre-authorized envelope (FD-EV §4.3).

Action Category	Consequence of Erroneous	Basis for Below-Bright-Line Placement
		Reversible. No cascading potential.
Regulation service deployment within pre-committed regulation capacity	Regulation performance degradation. ACE control quality reduction. Correctable in subsequent intervals.	The pre-committed regulation capacity bounds the consequence: an erroneous deployment cannot move more than the committed megawatts, and that bound is enforced by the AGC limits rather than by the recommendation. Reversible in the next interval. No cascading potential. The placement rests on that bound, not on the commitment having been authorized earlier.
Near-term interchange schedule adjustment within the operator's bilateral authority	Minor imbalance settlement exposure. Correctable in subsequent scheduling intervals.	The consequence is financial and within the operator's existing authority under the bilateral agreement—no reliability effect.

3.2 Proposed Above-Bright-Line Actions

The following action categories are proposed as above the Bright Line. For each, a CB confirmation with the disclosure element specified in Section 4.3 is required before the system executes the action. The CB confirmation is integrated into the operational interface as described in Section 4.

Action Category	Consequence of Erroneous	Basis for Above-Bright-Line Placement
Emergency energy request to neighboring BA	Erroneous requests may commit neighboring BA resources, affect their ACE, and trigger settlement obligations that cannot be recalled.	Cross-BA effect. Action is not reversible within the real-time horizon once acknowledged by the neighboring BA. Consequence extends beyond the operator's footprint.

Action Category	Consequence of Erroneous	Basis for Above-Bright-Line Placement
Demand response activation (involuntary load reduction)	Direct public impact through loss of electrical service to affected customers. Regulatory reporting obligation triggered.	Public impact. Involuntary customer action. Regulatory consequence. Not reversible within the response timeframe once the activation signal is sent.
Transmission switching or topology reconfiguration	Potential for overloads on adjacent paths, contingency exposure increase, and system state changes that are not fully predictable under current conditions.	System-wide effect. Consequences may not be immediately observable. Reversal requires a second switching operation with its own risk profile.
Dispatch instructions to a nuclear generating unit outside its pre-agreed maneuvering band, or toward its technical-specification limits	Technical specifications constrain nuclear unit dispatch. An erroneous dispatch instruction approaching technical specification limits may challenge reactor protection systems or require operator protective action.	An erroneous instruction of this kind can challenge technical-specification limits or reactor protection systems, a consequence above the threshold stated at the head of this section. The placement follows that consequence, not the unit's fuel type; incremental dispatch inside the band is placed below the line in Section 3.1. Confidence in the advisory output does not change the placement: these instructions warrant independent human evaluation, consistent with defense in depth.
Remedial Action Scheme (RAS) arm/disarm instruction	RAS status affects automatic system response to contingencies. An incorrect RAS status at the time of a contingency may trigger automatic actions inconsistent with current system conditions.	Pre-contingency consequence. The error is not visible until a contingency occurs. At that point, the consequence is immediate and potentially cascading.
Load shedding instructions under emergency conditions.	Highest-consequence action in the BA operator's toolkit. Direct, immediate public impact at a	Maximum public consequence. Irreversible within the response window.

Action Category	Consequence of Erroneous	Basis for Above-Bright-Line Placement
	significant scale. Regulatory reporting. Potential cascading effect on neighboring BAs if executed incorrectly.	Cascade potential. Explicit NERC standard obligations attach to this action category.

3.3 The Time-Critical Action Question

A specific challenge in the BA environment is the category of action that is above the Bright Line on consequence grounds but may develop on a timescale that severely compresses the CB evaluation window. A frequency excursion that develops over seconds toward under-frequency load-shedding thresholds is the canonical example.

The proposed resolution is not to move these actions below the Bright Line, but to recognize that the CB interface must be designed for rapid evaluation, not absent evaluation. Where the timescale compresses the evaluation window below what in-the-moment confirmation can meet, the conforming route is not reclassification but advance discharge of the §4.5 authorization by a verified deterministic gatekeeper under FD-BL-D1 (R1–R3) — the action remaining above the Bright Line — with in-the-moment Command Broker confirmation the fallback wherever the R2 completeness proof cannot be produced. A CB confirmation that takes three seconds in a well-designed integrated interface is operationally compatible with a frequency excursion response. A CB confirmation that requires the operator to navigate to a separate form, read a disclosure document, and execute a multi-step sign-off is not. The interface specification in Section 4 is designed to support genuine CB evaluation at the pace required by the operational environment.

Actions that are genuinely incompatible with any CB evaluation step — where the response window is too compressed for even a rapid integrated confirmation to be meaningful — should be examined to determine whether their automation should be handled by deterministic protective relay logic and RAS schemes rather than by Gen AI advisory output. If an action must execute faster than a human can meaningfully confirm it, that is a design signal that the action belongs in the deterministic protective system layer rather than in the Gen AI advisory layer. The Gen AI system should not be presenting options that the operator cannot evaluate. Moving an action to that layer does not move it below the Bright Line. Its consequence is unchanged, so it remains above the line, and the deterministic protective scheme becomes the verified gatekeeper that discharges the authorization in advance under FD-BL-D1, as the preceding paragraph states.

This proposed analysis is subject to review by BA operational subject matter experts. The author does not have direct operational BA experience and recognizes that the tempo characterization of specific action categories may require adjustment based on actual operational experience.

4. Interface Specification

This section specifies the minimum requirements for a CB-compliant interface in the BA deployment context. Requirements are stated as “MUST” (mandatory), “SHOULD” (recommended), or “MAY” (permitted). A system that satisfies all MUST requirements is CB-compliant for this use case. SHOULD requirements represent best practice.

4.1 General Display Requirements

R-GEN-1. The interface MUST present available action options in a single operational view. The operator MUST NOT be required to navigate away from the primary operational display to evaluate or confirm any action option, whether above or below the Bright Line.

R-GEN-2. The interface MUST display, for each action option, the current system state information on which the Gen AI recommendation is based. At minimum: current ACE value and trend, current frequency, the resource or path affected by the recommended action, and the expected ACE correction the Gen AI system projects for that action.

R-GEN-3. The interface MUST clearly distinguish above-Bright-Line options from below-Bright-Line options using a persistent visual treatment that is immediately apparent without requiring the operator to read option text. Color coding, iconography, or panel separation are acceptable treatments. Subtle differentiation — a small icon or minor color variation — does not satisfy this requirement.

R-GEN-4. The interface MUST display a validity timestamp for each presented option, indicating when the recommendation was generated. Options MUST be visually flagged as stale when the system state has changed materially since the recommendation was generated. The threshold for staleness flagging is defined in the system’s configuration parameters and must be documented.

R-GEN-5. The interface MUST allow the operator to act independently of the presented options at any time. The Gen AI advisory display MUST NOT occupy or obscure the primary system monitoring display elements that the operator requires for independent situational awareness.

4.2 Below-Bright-Line Option Display and Confirmation

Below-Bright-Line options require a simplified display and a positive confirmation gesture. They do not require the CB disclosure element.

R-BL-1. For each of the Bright Line options below, the interface must display: the action description in plain operational language; the sequence of system actions that will execute upon confirmation; the expected operational outcome; and any constraints or limits that bound the action sequence.

R-BL-2. The confirmation gesture for a below-Bright-Line option must be a deliberate two-step action: select the option, then enter a distinct confirmation input. A single button press or touch that both selects and confirms does not satisfy this requirement.

R-BL-3. When a below-Bright-Line option is confirmed, the system must display the action sequence being executed and provide real-time feedback as each action completes.

R-BL-4. The system MUST generate an audit record at the moment of the below-Bright-Line confirmation as specified in Section 5.

4.3 Above-Bright-Line Option Display and CB Confirmation

Above-Bright-Line options require the CB disclosure element integrated into the option display panel. The disclosure element is not a separate screen or document. It is a defined set of information fields displayed within the option panel, against which the operator performs their CB evaluation before confirming.

R-AL-1. For each above-Bright-Line option, the interface MUST display all below-Bright-Line information elements (R-BL-1) plus the following CB disclosure fields:

- Consequence characterization. A plain-language statement of the consequence category of this action, referencing the Bright Line placement basis from Section 3. Example: “This action initiates involuntary load reduction. Public impact. Regulatory reporting required.” The consequence characterization is system-generated and must reflect the placement basis documented in Section 3 of this use case.
- Anomaly flags. Any conditions identified by the Gen AI system that are anomalous relative to the system state used to generate this recommendation. If no anomalies are identified, the field must state “No anomalies identified” explicitly — a blank field does not satisfy this requirement.
- Confidence and basis summary. The Gen AI system’s characterization of the analytical basis for this recommendation is stated in terms that the operator can evaluate against their own situational awareness. This is not a confidence percentage — it is a plain-language statement of what the system is relying on to make this recommendation.
- Action boundary statement. An explicit statement of what this authorization covers: this specific action, at this magnitude, affecting these specific resources or paths, under the current system state. The operator is authorizing this action as described, not a class of similar actions.

R-AL-2. The CB confirmation gesture for an above-Bright-Line option MUST be a three-step action: selection of the option; a deliberate review pause — a minimum display time of the CB disclosure fields before the confirmation input is available, configurable by the deploying organization but not less than five seconds; and a distinct confirmation input that is physically

and visually separated from the selection input. The confirmation input MUST be labeled with language that references the CB obligation, such as “Authorize — CB Review Complete”.

R-AL-3. The minimum display time required by R-AL-2 MUST be suspendable by the operator in an identified emergency condition. The operator MUST provide an explicit emergency override input that is distinct from the normal confirmation input. Emergency override confirmation MUST generate a distinct audit record flag as specified in Section 5. The emergency override capability does not reduce the operator’s CB accountability; it provides a documented pathway for time-critical situations while preserving the audit record.

R-AL-4. When an above-Bright-Line option is confirmed, the system MUST NOT execute the action until the audit record entry is successfully written. Action execution is contingent on the integrity of audit records.

R-AL-5. The system MUST allow the operator to annotate an above-Bright-Line confirmation with a brief free-text note. This field is optional for the operator, but must be available. Operator annotations are incorporated into the audit record.

4.4 Mixed-Option Screen Behavior

The operational screen will frequently present above- and below-Bright-Line options simultaneously. The following requirements govern this condition.

R-MIX-1. Confirmation of a below-Bright-Line option MUST NOT remove or alter the display of pending above-Bright-Line options. Both tiers MUST remain independently actionable throughout the operational session.

R-MIX-2. When a below-Bright-Line action sequence is executing, the interface MUST continue to display above-Bright-Line options with their full CB disclosure elements. The operator MUST be able to initiate a CB confirmation of an above-Bright-Line option while a below-Bright-Line sequence is in progress.

R-MIX-3. When system conditions change materially while an above-Bright-Line option is displayed but not yet confirmed, the interface MUST update the CB disclosure fields to reflect current conditions. If the update changes the consequence characterization or anomaly flag status of the option, the review pause timer (R-AL-2) MUST reset.

R-MIX-4. An above-Bright-Line option that has become stale (R-GEN-4) must be visually flagged, and its confirmation input must be disabled until the operator explicitly acknowledges the staleness flag. The staleness acknowledgment generates an audit record entry distinct from the confirmation record.

4.5 Operator Independence Preservation

R-IND-1. The Gen AI advisory display MUST NOT be the only source of system state information available to the operator. Independent system state monitoring — ACE, frequency,

interchange, generation output — MUST be continuously visible on the primary operational display and MUST NOT be removed or obscured by the advisory interface at any time.

R-IND-2. The operator MUST be able to dismiss, minimize, or override any Gen AI recommendation without consequence to system operations. Dismissal of a recommendation is not an audit event. Confirmation of a recommendation is.

R-IND-3. The system MUST NOT present Gen AI recommendations in a manner that implies more urgency than the operational situation warrants. Visual or auditory alarm treatments for Gen AI recommendations MUST be configurable and MUST default to a lower urgency level than primary system alarms.

5. Audit Record Specification

Every operator confirmation — whether above or below the Bright Line — generates an audit record. The audit record is a legal document. Requirements in this section are mandatory for all CB-compliant systems deployed in this use case context.

5.1 Below-Bright-Line Confirmation Record

The audit record for a below-Bright-Line confirmation MUST contain:

- Timestamp of confirmation to one-second precision.
- Operator identifier — the full name or system-linked identifier of the operator who confirmed. The identifier must be linked to the operator's verified identity in the system authentication record.
- Action sequence as presented to the operator at the moment of confirmation, including all resources, magnitudes, and paths involved.
- System state snapshot at the moment of confirmation: ACE value, frequency, interchange net, and any anomaly flags present in the advisory system at the time of confirmation.
- Gen AI system identifier: name and version of the advisory system that generated the recommendation.

5.2 Above-Bright-Line CB Confirmation Record

The audit record for an above-Bright-Line CB confirmation MUST contain all below-Bright-Line confirmation record elements plus:

- The complete CB disclosure fields as displayed to the operator at the moment of confirmation: consequence characterization, anomaly flags, confidence and basis summary, and action boundary statement.
- Duration of display before confirmation: the actual time elapsed between first display of the CB disclosure fields and the confirmation input, in seconds.
- Emergency override flag, if the minimum display time was suspended under R-AL-3. If an emergency override was used, the record must include the override input timestamp and clearly indicate that it is an emergency override confirmation.
- Operator annotation, if provided.
- Action execution confirmation: timestamp at which the system executed the authorized action and a success or failure status.
- Audit record write confirmation: the system's internal confirmation that the audit record was successfully written before action execution commenced, per R-AL-4.

5.3 Record Format and Preservation

R-AUD-1. Audit records MUST be written in a format that is both machine- and human-readable and does not require proprietary software. JSON with a documented schema, XML with a published schema, or an equivalent open format satisfies this requirement.

R-AUD-2. Audit records MUST be written to a storage system that is logically independent of the Gen AI advisory system. A record written in the same system that generated the recommendation does not satisfy the independence requirement. The record storage system must be accessible to the deploying organization's compliance and legal personnel independently of the advisory system vendor.

R-AUD-3. Audit records MUST be preserved for a minimum of seven years from the date of the confirmation event, or for the retention period specified in the applicable sector annex, whichever is longer.

R-AUD-4. The audit record system MUST provide a tamper-evident mechanism. Records MUST NOT be modifiable after writing. The system MUST log any access to audit records, including read access.

6. Conformance Assessment

A system claiming CB conformance for this use case must demonstrate compliance with all MUST requirements in Sections 4 and 5. Conformance is assessed by the deploying

organization, not by the vendor. Vendor claims of conformance are not a substitute for deploying an organizational assessment.

The conformance assessment must be documented and address each MUST requirement individually, explaining how each requirement is satisfied and identifying any requirements for which the implementation differs from the specified approach. Deviations must include an explanation of why the deviation provides equivalent or superior protection of the CB function.

The conformance assessment document is an audit artifact. It must be maintained for the life of the deployment and updated when the system is modified in ways that affect any MUST requirement.

The deploying organization is responsible for verifying that the qualified CB operators assigned to the system have been trained on the specific interface implementation, including the visual treatment of above- versus below-Bright-Line options, the confirmation gesture sequence, and the emergency override procedure. Interface training is a component of CB qualification maintenance under the CB-IB_Qualification_Standard. That training is delivered through the Joint Training Event specified in the CB-CT Commissioning & Joint Training module (§5).

7. Cross-Sector Adaptation Guidance

The interface architecture specified in Sections 4 and 5 is cross-sectoral. The BA scenario provides the illustrative context. Vendors adapting this specification for other high-tempo operational environments substitute sector-specific content in the following elements without changing the underlying architecture.

7.1 Natural Gas Transmission Control Room

Substitute for BA-specific elements:

- ACE and frequency with linepack deviation, compressor station throughput differential, and pipeline pressure at key measurement points.
- Resource classes with compressor unit dispatch, valve position changes, and interconnect flow adjustments.
- Bright Line placement analysis with consequence analysis derived from the DBA-GAS sector annex consequence thresholds: uncontrolled release events, electricity-gas interdependency cascade, and cross-border supply disruption.
- NERC reliability standard references with applicable PHMSA and TSA pipeline security directive requirements.

The high-tempo decision cadence, mixed above- and below-Bright-Line option presentation, and integrated CB disclosure element architecture apply without modification.

7.2 Water System Operations Center

Substitute for BA-specific elements:

- ACE and frequency with treatment process parameters: chemical dosing rates, flow rates, pressure zones, and reservoir levels.
- Resource classes with treatment unit dispatch, pump station adjustments, valve configurations, and chemical feed system changes.
- Bright Line placement analysis with consequence analysis derived from the DBA-W sector annex consequence thresholds: population-scale public health impact, wastewater containment loss, and loss of service to critical dependents.

The 2021 SCADA intrusion in Oldsmar, Florida, is the reference adversarial event for the water-sector stress scenario. The interface independence preservation requirements (R-IND-1 through R-IND-3) are directly applicable and were developed with this event class in mind.

7.3 Data Center Network Operations Center

Substitute for BA-specific elements:

- ACE and frequency with IT load metrics, cooling system performance parameters, power distribution unit loading, and UPS reserve status.
- Resource classes with workload migration decisions, cooling unit dispatch, generator start sequences, and connectivity path switching.
- Bright Line placement analysis with consequence analysis derived from the DBA-DC sector annex consequence thresholds, with particular attention to the concentration characteristic: actions that affect multiple hyperscale facilities simultaneously carry elevated consequence profiles not visible at the single-facility level.

The March–April 2026 Iranian drone strikes on AWS and Oracle facilities in the UAE are the reference adversarial events for the data center sector. The anomaly flag requirement (R-AL-1, second bullet) must include flags for indicators consistent with pre-positioning or coordinated physical and cyber attack, as specified in the DBA-DC-L3_Pre_Positioning scenario.

8. Relationship to Parent Documents

This Document	Parent Requirement	Parent Document and Section
Section 3 — Bright Line placement	Bright Line was placed by consequence analysis, not system sophistication (governing definition FD-	CB-Framework_Cross_Sector, Section 3.2

This Document	Parent Requirement	Parent Document and Section
	BL — placement is consequence, per action; cited by reference)	
Section 4.3 — CB disclosure fields	CB must have access to the information needed to evaluate independently	CB-IB_Qualification_Standard, Sections 3.3 and 4.1
R-AL-2 — Confirmation gesture and review pause	The confirmation step must be local, auditable, and not executable remotely.	CB-Framework_Cross_Sector, Section 3.1
R-AL-3 — Emergency override	CB qualification must include stress-condition performance	CB-IB_Qualification_Standard, Section 3.6 and 4.1
Section 4.3 — Anomaly flags	CB adversarial scenario awareness domain	CB-IB_Qualification_Standard, Section 4.1, fifth domain
Section 5 — Audit records	Authorization records must be local, auditable, and tamper-evident	CB-Framework_Cross_Sector, Section 3.1; CB-IB_Qualification_Standard, Section 7
R-IND-1 through R-IND-3 — Operator independence	CB must have independent access to process data	CB-IB_Qualification_Standard, Section 4.1, third domain

9. References

Command Broker family instruments are cited in this document by designation, without version tokens. The current version of each is stated in CB-REG, which is the authoritative register for this family.

CB-Framework_Cross_Sector. Roxey, T. Eclectic Technologies, 2026.

CB-IB_Qualification_Standard. Roxey, T. Eclectic Technologies, 2026.

CB-CT_Commissioning_and_Joint_Training. Roxey, T.E. Eclectic Technologies, 2026.

NERC Reliability Standard BAL-001-2. Balance Resources and Demands. North American Electric Reliability Corporation.

NERC Reliability Standard EOP-006-3. System Restoration Coordination. North American Electric Reliability Corporation.

Roxey, T.E. Generative AI Risk in Industrial Control Systems: A Study for Critical Infrastructure Operators, Regulators, and Suppliers. Version 29. Eclectic Technologies, 2026.

DBA-DC-L3_Pre_Positioning v1.1. Roxey, T. Eclectic Technologies. [Reference for pre-positioning anomaly flag content in Section 7.3.]

Revision History

Version	Date	Description
0.1	May 2026	Initial draft. Establishes the BA operational scenario, proposed Bright Line placement with analytical basis for review, integrated interface specification with above/below Bright Line display and confirmation requirements, audit record specification, conformance assessment requirements, cross-sector adaptation guidance for gas transmission, water, and data center environments, and traceability to parent documents. Bright Line placement in Section 3 is proposed for review by BA operational subject matter experts.
0.1 (housekeeping)	June 2026	Housekeeping correction. “Brightline” (one word) corrected to “Bright Line” (two words, title case) throughout all body text and section headings, consistent with series-wide convention established in DBA-EN-SF §4.3. Compound forms “Above-Brightline” and “Below-Brightline” corrected to “Above-Bright-Line” and “Below-Bright-Line” accordingly. No substantive content changes.
0.1 (housekeeping)	July 2026	Housekeeping correction. Source-study citation in the References section updated to the canonical edition — Roxey, T.E., “Generative AI Risk in Industrial Control Systems,” Version 29, Eclectic Technologies (2026) — replacing the prior ISAGCA-attributed “A Study of AI Risks — Vendor Agnostic,” Version 28. Citation entry only; no substantive content changes.
0.1 (cite refresh)	Jul 2026	Citations refreshed to current corpus versions: CB-Framework v0.2, Command Broker Implementation Guide V0.2, CB-CT v0.2, DBA-VC v0.8. Mechanical version-reference pass; no substantive content change. (Held pending decision: the NERC/BES-Variant TAF reference and the ISA Parent Framework V1.2 label.)
0.2	September 2026	Citation-strip pass, version-pin pass Tier 4, and the document's first version increment: the three preceding entries record revisions taken without one. Command Broker family citations taken to the family register form. Fifteen version tokens stripped — the cover masthead, the parent-document reference line, seven cells of the Section 8 parent-requirement mapping table, and three Section 9

Version	Date	Description
		reference entries — and the CB-REG lead-in added at the head of Section 9. The FD-BL pin in the Section 8 mapping table stripped under the same convention, the Foundational Definitions being cited by designation with FD-REG authoritative for their version state. The Section 9 entry for DBA-DC-L3_Pre_Positioning is left pinned: DBA-EN-SF Section 1.4 registers the Critical Loads grouping as a single row and does not carry that instrument's version state, so a stripped citation would resolve to nothing. The pin is current as it stands.
0.3	October 2026	Placement conformance to FD-BL §4.2, in answer to Project 2's note of 30 September 2026, which asked whether Section 3 is clean before using this use case as the NERC worked exemplar in Governing Generative AI in Critical Infrastructure Control. Section 3 already placed each action on its consequence; four passages are tightened. The opening of Section 3 states the per-action consequence criterion and cites FD-BL §4.2. The battery and regulation rows in Section 3.1 rest on a consequence bound enforced outside the recommendation, and the regulation row no longer rests on the commitment having been authorized earlier, which is a mode-of-discharge argument rather than a placement. The nuclear dispatch row is split by consequence on the author's ruling of 1 October 2026: incremental dispatch inside the unit's pre-agreed maneuvering band is placed below the line in Section 3.1, and instructions outside the band or toward technical-specification limits stay above it in Section 3.2, with the threshold at the head of Section 3 extended to name that consequence. Section 3.3 states that an action moved to the deterministic protective layer remains above the line and is discharged by that layer as an FD-BL-D1 gatekeeper. Section 3.1's opening is corrected from a single confirmation gesture to the two-step confirmation that R-BL-2 requires.
0.4	October 2026	Conformed to the October 2026 Edition of the Foundational Definitions, on Project 0's work order of 3 October 2026 (drift-screen finding P8-06) and Project 2's notice of the same day. The battery-dispatch row in Section 3.1 placed the action below the line on the right ground, the consequence bound enforced by the battery controller's limits, but named that bound with the reserved term "pre-authorized envelope," which a reader could take as placement by envelope. The row now names the bound as the controller's operating limits, states that a below-line placement needs no authorization, and cites FD-EV §4.3 for the operating-band sense. This matches the reading CB-PAE §2.1 now takes of this row. No placement changes. The cover date, which still read May 2026, is corrected. Publication handling under the CR-PB draft: v0.3 is current on the publications site, so this pass acquires a republish step.